Sheet 29 Mon. 11 Jan. 2027 Grouped data: histogram and cumulative frequencies

1. Warm-up · Flash Give the width and the midpoint of the class $[20;35[$.
2. Core · Table Classes $[0;10[$, $[10;20[$, $[20;30[$ with frequencies 4, 6, 10. Give the relative frequencies then the increasing cumulative frequencies.
3. Core + · Flash In a histogram with classes of equal width, two rectangles have heights 3 cm and 6 cm. What is the ratio of their frequencies?
4. Back · Flash Write 0.00032 in scientific notation.
5. Running thread · Flash Solve in $\mathbb{R}$: $2(x+3) = 5x - 9$.

Sheet 30 Tue. 12 Jan. 2027 Weighted mean, median class; spreadsheet and Python

1. Warm-up · Flash Calculate $\dfrac{2\times 8 + 3\times 12}{5}$.
2. Core · Method Classes $[0;10[$, $[10;20[$, $[20;30[$ with frequencies 4; 7; 9. Estimate the mean using the midpoints classes; give the median class.
3. Core + · Reading code What does this program calculate and what value is obtained?

```
notes = [12, 15, 9, 14]
s = 0
for n in notes:
    s = s + n
print(s / len(notes))
```

4. Back · Flash A 45,000 Ar item is reduced to -30%. What is the sale price?
5. Running thread · Prove Procedure: choose n ; add 1 ; square ; subtract n^2. Show that the result is always an odd number.

Algebraic calculation, equations and inequalities

Sheet 31 Thu. 14 Jan. 2027 Integer powers (positive and negative)

1. Warm-up · Flash Calculate 2^{-3}, 10^{-2} and $(-3)^2$.
2. Core · Method Simplify $\dfrac{2^3 \times 2^{-5}}{2^{-1}}$.
3. Core + · Method For $x \neq 0$, write $(x^2)^{-3} \times x^7$ in the form x^n.
4. Back · Flash Median of the data set 4, 7, 9, 10, 12, 15?
5. Running thread · Recognise Which solid has a disc-shaped base, a curved lateral surface and a single vertex?

Sheet 32 Mon. 18 Jan. 2027 Square roots; $\sqrt{a^2} = |a|$

1. Warm-up · Flash Calculate $\sqrt{36}$ and $\sqrt{0{,}09}$.
2. Core · Method Write $\sqrt{72}$ in the form $a\sqrt{2}$ and calculate $\sqrt{(-5)^2}$.
3. Core + · True/false True or false? Justify. $\sqrt{9+16} = \sqrt{9} + \sqrt{16}$.
4. Back · Flash Calculate $(3^2)^3$.
5. Running thread · True/false True or false? Justify. The diagonals of a rhombus are perpendicular.

Sheet 33 Tue. 19 Jan. 2027 Algebraic fractions; expanded, simplified or factorised forms

1. Warm-up · Flash Calculate $\frac{1}{2}+\frac{1}{3}$.

2. Core · Method Write as a single fraction $\frac{1}{x}+\frac{1}{x+1}$ (for $x \neq 0$ and $x \neq -1$).

3. Core + · Method Factorise x^2-4 then simplify $\frac{x^2-4}{x-2}$ for $x \neq 2$.

4. Back · Flash Calculate $2a-3b$ for $a=2$ and $b=-1$.

5. Running thread · Flash ABCDEFGH is a cuboid with $AB=4$, $AD=3$, $AE=2$. In the coordinate system $(A;B,D,E)$, what are the coordinates of G?

Sheet 34 Thu. 21 Jan. 2027 Inequalities: sum, multiplication by a real number

1. Warm-up · Flash Complete with $<$ or $>$: $-3 \ldots -2$; $\frac{1}{3} \ldots 0{,}3$.

2. Core · Method We have $a<b$. Compare $-2a$ and $-2b$, then $a+5$ and $b+5$.

3. Core + · Method Given $2<x<3$ and $1<y<4$, bound $x+y$ then $x-y$.

4. Back · Flash Calculate $(-3)\times(-4)+2\times(-5)$.

5. Running thread · Problem A 120 cm by 84 cm sheet is cut into identical squares, with no waste. What is the largest possible side?

Sheet 35 Mon. 25 Jan. 2027 Comparison by difference and quotient

1. Warm-up · Flash Compare $\frac{2}{3}$ and $\frac{3}{4}$ by studying the sign of the difference.

2. Core · Method Compare x^2+1 and $2x$ for every real number x.

3. Core + · Method Compare $\frac{7}{5}$ and $\frac{8}{6}$ by quotient (positive numbers).

4. Back · Table Monthly precipitation (fictitious data).

Month	Jan.	Feb.	March	Apr.
Rain (mm)	280	300	240	120

Which month is the wettest? What is the total over these 4 months?

5. Running thread · Table → formula x : -2, -1, 0, 1 and $f(x)$: 4, 1, 0, 1. Is the function affine? Propose a formula.

Sheet 36 Tue. 26 Jan. 2027 Equations and inequalities of degree 1er degree; $x^2=a$; zero product

1. Warm-up · Flash Solve 3 $x-5=7$.

2. Core · Method Solve 2 $x+3 \leqslant 7-x$.

3. Core + · Method Solve $(x-2)(2x+6)=0$ then $x^2=7$.

4. Back · Flash Compare 1.4 and $\sqrt{2}$ without a calculator.

5. Running thread · Flash ABC is isosceles at A with $\widehat{BAC}=40°$. What is the measure of $\widehat{ABC}$?

Sheet 37 Thu. 28 Jan. 2027 Quotient equation $\frac{A(x)}{B(x)}=k$

1. Warm-up · Flash For which value of x the expression $\frac{1}{x-3}$ is it not defined?

2. Core · Method Solve $\frac{2x-1}{x+3}=0$.

3. Core + · Method Solve $\frac{x+1}{x-2}=3$.

4. Back · Expand Expand and simplify $(x+2)(x-5)$.